# THE RENORMALIZATION GROUP

## Developing the Formalism

For the $\overline{MS}$ renormalization scheme, we used the $\mu$ independence of physical observables to find the mass param and beta function.

It turns out that we can get these results from a more formal manner.

Equations that tell us how the lagrangian parameters vary with $\mu$ are called the equations of the renormalization group.

Let's write down the lagrangian of our theory:

$$\mathcal{L} = -\frac{1}{2} Z_\varphi \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} Z_m m^2 \varphi^2 + \frac{1}{6} Z_g g \tilde{\mu}^{\epsilon/2} \varphi^3 + Y\varphi \; ; \quad \mathcal{L} = -\frac{1}{2} \partial^\mu \varphi_0 \partial_\mu \varphi_0 - \frac{1}{2} m_0^2 \varphi_0^2 + \frac{1}{6} g_0 \varphi_0^3 + Y_0 \varphi_0$$

In the first eq$^n$, we are working with the renormalized fields and parameters. (Renormalized using the $\overline{MS}$ scheme) The next lagrangian is made of the bare fields and parameters:

$$\varphi_0(x) = Z_\varphi^{1/2} \varphi(x) \; ; \; m_0 = Z_\varphi^{-1/2} Z_m^{1/2} m \; ; \; g_0 = Z_\varphi^{-3/2} Z_g g \tilde{\mu}^{\epsilon/2} \; ; \; Y_0 = Z_\varphi^{-1/2} Y$$

Now, after using dimensional regularization, the loop infinities are inverse powers of $\epsilon$. In the $\overline{MS}$ scheme, we are using the Z's only to cancel these inverse powers. So we can write them as:

$$Z_\varphi = 1 + \sum_{n=1}^{\infty} \frac{a_n(\alpha)}{\epsilon^n} \; ; \; Z_m = 1 + \sum_{n=1}^{\infty} \frac{b_n(\alpha)}{\epsilon^n} \; ; \; Z_g = 1 + \sum_{n=1}^{\infty} \frac{c_n(\alpha)}{\epsilon^n}$$

Now, remember that we get the loop and vertex corrections in a taylor series in $\alpha$. So far we have:

$$a_1(\alpha) = -\frac{1}{6}\alpha + O(\alpha^2) \; , \quad b_1(\alpha) = -\alpha + O(\alpha^2) \; , \quad c_1(\alpha) = -\alpha + O(\alpha^2)$$

$\downarrow$

$a_1(\alpha), b_1(\alpha), c_1(\alpha)$ are atleast $O(\alpha^2)$ for $n \geq 2$

Now, with all of the obvious facts stated, let us try to compute some useful functions such as the beta function. The trick we use to do this is to set the bare parameters independent of $\mu$.

This is because since the exact scattering amplitudes are independent of $\mu$, the bare parameters must also be.

Now: $\alpha_0 \equiv \frac{g_0^2}{(4\pi)^3} = Z_g^2 Z_\varphi^{-3} \tilde{\mu}^\epsilon \alpha \; ; \quad G(\alpha, \epsilon) \equiv \ln(Z_g^2 Z_\varphi^{-3})$

From the general structure of the Z's, we have: $G(\alpha, \epsilon) = \sum_{n=1}^{\infty} \frac{G_n(\alpha)}{\epsilon^n}$

Now, $a_1(\alpha) = \ln\left(\left(\frac{c_0\alpha}{c}\right)^2\left(\frac{\epsilon}{a_1\alpha}\right)^3\right) = 2\ln\left(\frac{c_0\alpha}{\epsilon}\right) - 3\ln\left(\frac{a_1\alpha}{\epsilon}\right) = 2c_1(\alpha) - 3a_1(\alpha) = -\frac{3}{2} + O(\alpha^2)$

And: $\alpha_0 = Z_g^2 Z_\varphi^{-3}\hat{\mu}^\epsilon\alpha \longrightarrow \ln\alpha_0 = \ln(Z_g^2 Z_\varphi^{-3}\hat{\mu}^\epsilon\alpha) = a(\alpha,\epsilon) + \ln\alpha + \epsilon\ln\hat{\mu}$

Since $\alpha_0$ is a bare parameter, we need it to be invariant to $\mu$, so differentiating w.r.t $\ln\mu$:

$$0 = \frac{\partial a(\alpha,\epsilon)}{\partial\alpha}\frac{\partial\alpha}{\partial\ln\mu} + \frac{1}{\alpha}\frac{d\alpha}{d\ln\mu} + \epsilon = \left(1 + \alpha\frac{a_1'(\alpha)}{\epsilon} + \alpha\frac{a_2'(\alpha)}{\epsilon^2} + \dots\right)\frac{d\alpha}{d\ln\mu} + \epsilon$$

Now, if we think about it, $d\alpha/d\ln\mu$ is the rate at which $\alpha$ must change to compensate for a change in $\ln\mu$. This rate should be finite in the $\epsilon\to 0$ limit. So, in a renormalizable theory, we get:

$$\frac{d\alpha}{d\ln\mu} = -\epsilon\alpha + \beta(\alpha)$$

This finite term kicks in when the limit is $\epsilon\to 0$

Matching terms gives us: $\beta(\alpha) = \alpha^2 a_1'(\alpha)$

Q: How does this even happen?? As in the math behind the above eq$^n$.

Plugging the values in, we can see that this is the same beta-function as what we got in the $\overline{MS}$ scheme.

We can do the same for $m_0$. Start by defining:

$$m(\alpha,\epsilon) \equiv \ln(Z_m^{1/2} Z_\varphi^{-1/2}) = \sum_{n=1}^{\infty}\frac{m_n(\alpha)}{\epsilon^n}$$

Similar to how we calculated $a_1(\alpha)$, we have: $m_1(\alpha) = -\frac{5}{12}\alpha + O(\alpha^2)$

We have: $\ln m_0 = m(\alpha,\epsilon) + \ln m \longrightarrow \frac{\partial m(\alpha,\epsilon)}{\partial\alpha}\left(-\epsilon\alpha + \beta(\alpha)\right) + \frac{1}{m}\frac{dm}{d\ln\mu} = 0$

Re-arranging this gives us: $\frac{1}{m}\frac{dm}{d\ln\mu} = \alpha m_1'(\alpha) + \dots$

However, since we need our rate to be finite in the $\epsilon\to 0$ limit. The terms in the ellipses which are inverse orders of $\epsilon$ must actually be zero. This gives the anomalous dimension as:

$\gamma_m(\alpha) \equiv \frac{1}{m}\frac{dm}{d\ln\mu} = \alpha m_1'(\alpha) = -\frac{5}{12}\alpha + O(\alpha^2) \longrightarrow$ Same as our earlier results.

## Looking at Propagators

Consider the propagator in the $\overline{MS}$ renormalization scheme:

$$\tilde{\Delta}(k^2) = i\int d^6x\, e^{-ikx} \langle 0|T\varphi(x)\varphi(0)|0\rangle$$

The bare propagator is: $\tilde{\Delta}_0(k^2) = i\int d^6x\, e^{-ikx} \langle 0|T\varphi_0(x)\varphi_0(0)|0\rangle$ → must be $\mu$-independent.

These two are related by: $\tilde{\Delta}_0(k^2) = Z_\varphi \tilde{\Delta}(k^2)$

Doing the usual log and differentiating:

$$0 = \frac{d\ln Z_\varphi}{d\ln\mu} + \frac{1}{\tilde{\Delta}(k^2)}\left(\frac{\partial}{\partial\ln\mu} + \frac{d\alpha}{d\ln\mu}\frac{\partial}{\partial\alpha} + \frac{dm}{d\ln\mu}\frac{\partial}{\partial m}\right)\tilde{\Delta}(k^2)$$

If we write: $\ln Z_\varphi = \frac{a_1(\alpha)}{\epsilon} + \frac{a_2(\alpha) - \frac{1}{2}a_1^2(\alpha)}{\epsilon^2} + \dots$

This gives us: $\frac{d\ln Z_\varphi}{d\ln\mu} = \frac{\partial\ln Z_\varphi}{\partial\alpha}\frac{d\alpha}{d\ln\mu} = \left(\frac{a_1'(\alpha)}{\epsilon} + \dots\right)\left(-\epsilon\alpha + \beta(\alpha)\right) = -\alpha a_1'(\alpha) + \dots$

The ellipses must zero out since the propagator is finite in the $\epsilon \to 0$ limit.

We then define the anomalous dimension of the field:

$$r_\varphi(\alpha) \equiv \frac{1}{2}\frac{d\ln Z_\varphi}{d\ln\mu} = -\frac{1}{2}\alpha a_1'(\alpha) = +\frac{1}{12}\alpha + O(\alpha^2)$$

This gives us:

$$\left(\frac{\partial}{\partial\ln\mu} + \beta(\alpha)\frac{\partial}{\partial\alpha} + r_m(\alpha)\, m\frac{\partial}{\partial m} + 2r_\varphi(\alpha)\right)\tilde{\Delta}(k^2) = 0$$

↑

I think, we can picture this equation as: the changes to the propagator due to changing $\mu$ are compensated by changes due to $\alpha, m$ and the field through its anomalous dimension. Thus this equation keeps the physics accurate.

This equation is called the Callan-Symanzik equation for the propagator.